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Method of identifying target device based on reception of utterance and electronic device therefor

Abstract

There is disclosed an electronic device including a communication circuitry, a processor, and a memory that stores instructions. The electronic device may be configured to acquire, from a plurality of external electronic devices that have received a wake-up utterance, state information indicating states of the plurality of external electronic devices, acquire control data corresponding to a control utterance subsequent to the wake-up utterance and instructing control of a first function, determine availability of the first function for at least one of the plurality of external electronic devices by using the state information, determine a target device to perform the first function from the plurality of external electronic devices, based on the availability, and control the target device so as to perform the first function.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation of International Application No. PCT/KR2022/014154, which was filed on Sep. 22, 2022, and claims priority to Korean Patent Application No. 10-2021-0144994, filed on Oct. 27, 2021, in the Korean Intellectual Property Office, the disclosure of which are incorporated by reference herein their entirety.

BACKGROUND

Technical Field

(1) One or more embodiments of the present disclosure generally relate to a method of identifying a target device based on reception of an utterance and an electronic device therefor.

Description of Related Art

(2) Techniques for controlling an electronic device based on voice commands of a user are being widely used. For example, the electronic device may include a voice assistant application or program configured to identify the user's intent from the user's utterance and perform an action corresponding to the identified intent. The user may control the electronic device through these voice commands. With the increasing number of electronic devices supporting the voice assistance, the user may control various devices based on speech. For example, the user may activate the voice assistant by uttering a wake-up word. After activation of the voice assistant, the user may control a specific function of the electronic device by using a control utterance instructing control of the specific function.

SUMMARY

(3) With the increasing number of electronic devices supporting voice assistance, a plurality of electronic devices adjacent to the user may all be configured to support voice assistance. In this case, a plurality of electronic devices may respond to a specified utterance of the user (e.g., wake-up word), including the device intended by the user and other devices not intended by the user. When a plurality of electronic devices respond to the utterance of the user, the user may have difficulty controlling the intended electronic device via voice command.

(4) In order to prevent the plurality of electronic devices from simultaneously responding, only one winner device in the plurality of electronic devices receiving the wake-up utterance may be designated to provide a response to the user. However, experience of the user may vary depending on which device is determined as the winner device. For example, the intent of the user may be to increase the volume of the TV. If the TV is selected as the winner device, the user may increase the volume of the TV by simply uttering "volume up". On the other hand, if a device other than the TV is selected as the winner device, the user may have to indicate a target device by uttering "turn up the TV volume". As described above, the utterance of the user may have to be changed depending on the determination of the winner device, which may inconvenience the user.

(5) In another example, the user may be forced to always include the name of the device in control utterances. However, in this case, because the user has to include the name of the target device in the control utterances, the user is also inconvenienced. Furthermore, as the number of devices to be controlled increases, identification of a target device may become more difficult.

- (6) Certain embodiments of the present disclosure may provide an electronic device and a method for solving the above-described problems.
- (7) According to an aspect of the present disclosure, there is provided an electronic device including a communication circuitry, a processor, and a memory that stores instructions, and the instructions, when executed by the processor, cause the electronic device to acquire, from a plurality of external electronic devices that have received a wake-up utterance, state information indicating states of the plurality of external electronic devices, acquire control data corresponding to a control utterance subsequent to the wake-up utterance, the control utterance instructing control of a first function, determine availability of the first function for at least one of the plurality of external electronic devices by using the state information, determine a target device to perform the first function from the plurality of external electronic devices, based on the availability, and control the target device so as to perform the first function.
- (8) According to another aspect of the present disclosure, there is provided a method for controlling a target device of an electronic device, the method including acquiring, from a plurality of external electronic devices that have received a wake-up utterance, state information indicating states of the plurality of external electronic devices, acquiring control data corresponding to a control utterance subsequent to the wake-up utterance, the control utterance instructing control of a first function, determining availability of the first function for at least one of the plurality of external electronic devices by using the state information, determining a target device to perform the first function from the plurality of external electronic devices, based on the availability, and controlling the target device so as to perform the first function.
- (9) Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The above and other aspects, features, and advantages of certain embodiments of the present disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:
- (2) FIG. 1 is a block diagram illustrating an electronic device in a network environment according to various embodiments.
- (3) FIG. 2 is a block diagram illustrating an integrated intelligence system according to an embodiment.
- (4) FIG. 3 is a diagram illustrating a form in which information on relation between concepts and actions is stored in a database, according to an embodiment.
- (5) FIG. 4 is a diagram illustrating a user terminal displaying a screen for processing a voice input received through an intelligent app, according to an embodiment.
- (6) FIG. 5 illustrates a system for controlling a target device based on an utterance, according to an example.
- (7) FIG. 6 illustrates a multi-device environment according to an example.
- (8) FIG. 7 illustrates a block diagram of an electronic device according to an example.
- (9) FIG. 8 illustrates a system for controlling a listener device according to an example.
- (10) FIG. 9 is a signal flow diagram illustrating a method for controlling a target device according to an example.
- (11) FIG. 10 is a signal flow diagram illustrating a method for controlling a target device according to an example.
- (12) FIG. 11 illustrates a control environment of a target device according to an example.
- (13) FIG. 12 is a flowchart illustrating a method for controlling a target device according to an

example.

DETAILED DESCRIPTION

(14) The system according to an embodiment of the present disclosure may control an electronic device according to the intent of the utterance of a user, thereby improving user convenience and utility of the electronic device.

(15) An electronic device according to an example of the present disclosure may identify a target device in consideration of the intent of the user and states of various electronic devices, thereby improving user convenience, which may drive the user to use the electronic devices more frequently.

(16) The electronic device according to an example of the present disclosure may use utterance data and the state of an electronic device together, thereby reducing input steps of the user.

(17) Hereinafter, various embodiments disclosed in the present disclosure will be described with reference to the accompanying drawings. However, this is not intended to limit the present disclosure to the specific embodiments, and it is to be construed to include various modifications, equivalents, and/or alternatives of embodiments of the present disclosure.

(18) FIG. 1 is a block diagram illustrating an electronic device **101** in a network environment **100** according to various embodiments. Referring to FIG. 1, the electronic device **101** in the network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or at least one of an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some embodiments, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some embodiments, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**).

(19) The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. According to one embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

(20) The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor

121 while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

(21) The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

(22) The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

(23) The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

(24) The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

(25) The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

(26) The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

(27) The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

(28) The interface **177** may support one or more specified protocols to be used for the electronic

device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

(29) A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

(30) The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

(31) The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

(32) The power management module **188** may manage power supplied to the electronic device **101**. According to one embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

(33) The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

(34) The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

(35) The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may

support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

(36) The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module **197** may include an antenna including a radiating element composed of a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

(37) According to various embodiments, the antenna module **197** may form a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

(38) At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

(39) According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In another embodiment, the external electronic device **104** may include

an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

(40) The electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

(41) It should be appreciated that various embodiments of the present disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

(42) As used in connection with various embodiments of the disclosure, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

(43) Various embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., internal memory **136** or external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

(44) According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application

store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

(45) According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to various embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

(46) FIG. 2 is a block diagram illustrating an integrated intelligence system according to an embodiment.

(47) Referring to FIG. 2, the integrated intelligent system according to an embodiment may include a user terminal **201**, an intelligent server **300**, and a service server **400**.

(48) The user terminal **201** (e.g., the electronic device **101** of FIG. 1) according to an embodiment may be a terminal device (or electronic device) connectable to the Internet, for example, a mobile phone, a smartphone, a personal digital assistant (PDA), a laptop computer, a television (TV), a white home appliance, a wearable device, a head mounted device (HMD), or a smart speaker.

(49) According to the illustrated embodiment, the user terminal **201** may include a communication interface **290**, a microphone **270**, a speaker **255**, a display **260**, a memory **230**, and/or a processor **220**. The components listed above may be operatively or electrically connected to each other.

(50) The communication interface **290** (e.g., the communication module **190** of FIG. 1) may be configured to be connected to an external device to transmit/receive data. The microphone **270** (e.g., the audio module **170** of FIG. 1) may receive sound (e.g., an utterance of the user) and convert the sound into an electrical signal. The speaker **255** (e.g., the sound output module **155** of FIG. 1) may output the electrical signal as sound (e.g., voice). The display **260** (e.g., the display module **160** of FIG. 1) may be configured to display images or videos. The display **260** according to an embodiment may also display a graphic user interface (GUI) of an executed app (or an application program).

(51) The memory **230** (e.g., the memory **130** of FIG. 1) according to an embodiment may store a client module **231**, a software development kit (SDK) **233**, and a plurality of applications. The client module **231** and the SDK **233** may constitute a framework (or solution program) for performing various functions. In addition, the client module **231** or the SDK **233** may constitute a framework for processing voice input.

(52) The plurality of applications (e.g., **235a** and **235b**) may be programs for performing specified functions. According to an embodiment, the plurality of applications may include a first app **235a** and/or a second app **235b**. According to an embodiment, each of the plurality of applications may include instructions for a plurality of operations that together performing a specified function. For example, the applications may include an alarm app, a message app, and/or a schedule app.

According to an embodiment, the plurality of applications may be executed by the processor **220** to sequentially execute at least some of the plurality of operations.

(53) The processor **220** according to an embodiment may control the overall operations of the user terminal **201**. For example, the processor **220** may be electrically connected to the communication interface **290**, the microphone **270**, the speaker **255**, and the display **260** to perform a specified

operation. For example, the processor **220** may include at least one processor. The processor **220** may include a microprocessor or any suitable type of processing circuitry, such as one or more general-purpose processors (e.g., ARM-based processors), a Digital Signal Processor (DSP), a Programmable Logic Device (PLD), an Application-Specific Integrated Circuit (ASIC), a Field-Programmable Gate Array (FPGA), a Graphical Processing Unit (GPU), a video card controller, etc. In addition, it would be recognized that when a general purpose computer accesses code for implementing the processing shown herein, the execution of the code transforms the general purpose computer into a special purpose computer for executing the processing shown herein. Certain of the functions and steps provided in the Figures may be implemented in hardware, software or a combination of both and may be performed in whole or in part within the programmed instructions of a computer. No claim element herein is to be construed as means-plus-function, unless the element is expressly recited using the phrase “means for.” In addition, an artisan understands and appreciates that a “processor” or “microprocessor” may be hardware in the claimed disclosure.

(54) The processor **220** according to an embodiment may also execute a program stored in the memory **230** to perform a specified function. For example, the processor **220** may execute at least one of the client module **231** and the SDK **233** to perform the following operations for processing voice input. The processor **220** may control operations of a plurality of applications through, for example, the SDK **233**. The following operations described as operations of the client module **231** or SDK **233** may be operations performed by execution of the processor **220**.

(55) The client module **231** according to an embodiment may receive a voice input. For example, the client module **231** may receive a voice signal corresponding to an utterance of the user detected through the microphone **270**. The client module **231** may transmit the received voice input (e.g., voice signal) to the intelligent server **300**. The client module **231** may transmit, to the intelligent server **300**, state information about the user terminal **201** together with the received voice input. The state information may be, for example, execution state information for an app.

(56) The client module **231** according to an embodiment may receive a result corresponding to the received voice input from the intelligent server **300**. For example, if the intelligent server **300** calculates a result corresponding to the received voice input, the client module **231** may receive the result corresponding to the received voice input. The client module **231** may display the received result on the display **260**.

(57) The client module **231** according to an embodiment may receive a plan corresponding to the received voice input. The client module **231** may display, on the display **260**, execution results of a plurality of actions of the app executed according to the plan. The client module **231** may, for example, sequentially display, on the display **260**, the execution results of the plurality of actions. In another example, the user terminal **201** may display only some execution results of the plurality of actions (e.g., the result of the last action) on the display **260**.

(58) According to an embodiment, the client module **231** may receive a request for obtaining information necessary for calculating the result corresponding to the voice input from the intelligent server **300**. According to an embodiment, the client module **231** may transmit the necessary information to the intelligent server **300** in response to the request.

(59) The client module **231** according to an embodiment may transmit, to the intelligent server **300**, result information obtained by executing the plurality of actions according to the plan. The intelligent server **300** may confirm that the voice input received by using the result information has been correctly processed.

(60) The client module **231** according to an embodiment may include a speech recognition module. According to an embodiment, the client module **231** may recognize a voice input to perform a limited function by using the speech recognition module. For example, the client module **231** may execute an intelligent app for processing a specified voice input (e.g., wake up!) by performing sequential operations in response to the voice input.

(61) The intelligent server **300** according to an embodiment may receive information related to the voice input of the user from the user terminal **201** through a network **299** (e.g., the first network **198** and/or the second network **199** of FIG. 1). According to an embodiment, the intelligent server **300** may convert data related to the received voice input into text data (e.g. by performing speech-to-text recognition). According to an embodiment, the intelligent server **300** may generate at least one plan for performing the task corresponding to the voice input of the user based on the text data.

(62) According to one embodiment, the plan may be generated by an artificial intelligent (AI) system. The artificial intelligence system may be a rule-based system, and may be a neural network-based system (e.g., feedforward neural network (FNN), and/or recurrent neural network (RNN)). Alternatively, the artificial intelligence system may be a combination of those described above, or another artificial intelligence system other than those described above. According to an embodiment, the plan may be selected from a set of predefined plans or may be generated in real time in response to the user request corresponding to the voice input. For example, the artificial intelligence system may select at least one plan from among a plurality of predefined plans.

(63) The intelligent server **300** according to an embodiment may transmit the result according to the generated plan to the user terminal **201** or transmit the generated plan to the user terminal **201**. According to an embodiment, the user terminal **201** may display the result according to the plan on the display **260**. According to an embodiment, the user terminal **201** may display, on the display **260**, the result obtained by executing actions according to the plan.

(64) The intelligent server **300** according to an embodiment may include a front end **310**, a natural language platform **320**, a capsule database **330**, an execution engine **340**, an end user interface **350**, a management platform **360**, a big data platform **370**, or an analytic platform **380**.

(65) The front end **310** according to an embodiment may receive the voice input received by the user terminal **201** from the user terminal **201**. The front end **310** may transmit a response corresponding to the voice input to the user terminal **201**.

(66) According to an embodiment, the natural language platform **320** may include an automatic speech recognition module (ASR module) **321**, a natural language understanding module (NLU module) **323**, a planner module **325**, a natural language generator module (NLG module) **327**, and/or a text-to-speech module (TTS module) **329**.

(67) The automatic speech recognition module **321** according to an embodiment may convert the voice input received from the user terminal **201** into text data. The natural language understanding module **323** according to an embodiment may determine an intent of the user by using text data of the voice input. For example, the natural language understanding module **323** may determine the intent of the user by performing syntactic analysis and/or semantic analysis. The natural language understanding module **323** according to an embodiment may identify the meaning of words by using linguistic features (e.g., grammatical elements) of morphemes or phrases, and determine the intent of the user by matching the meaning of the identified word with the intent.

(68) The planner module **325** according to an embodiment may generate a plan by using the intent and parameters determined by the natural language understanding module **323**. According to an embodiment, the planner module **325** may determine a plurality of domains required to perform a task required by the voice input based on the determined intent. The planner module **325** may determine a plurality of actions in each of the plurality of domains determined based on the intent. According to an embodiment, the planner module **325** may determine parameters required to execute the determined plurality of actions or the result value(s) outputted by the execution of the plurality of actions. The parameter and the result value(s) may be defined in a specified format (or class). Accordingly, the plan may include a plurality of actions and/or a plurality of concepts determined by the intent of the user. The planner module **325** may determine the relationship between the plurality of actions and the plurality of concepts in stages (or hierarchically). For example, the planner module **325** may determine the execution order of the plurality of actions determined based on the intent of the user based on the plurality of concepts. In other words, the

planner module **325** may determine the execution order of the plurality of actions based on parameters required for execution of the plurality of actions and results output by the execution of the plurality of actions. Accordingly, the planner module **325** may generate a plan including information (e.g., ontology) on the relation between the plurality of actions and the plurality of concepts. The planner module **325** may generate the plan by using information stored in the capsule database **330** in which sets of relationships between concepts and actions are stored.

(69) The natural language generator module **327** according to an embodiment may change specified information into text format. The information changed to the text format may be in the form of natural language utterance. The text-to-speech module **329** according to an embodiment may change information in the text format into information in voice format.

(70) According to an embodiment, some or all of the functions of the natural language platform **320** may be implemented in the user terminal **201** as well. For example, the user terminal **201** may include an automatic speech recognition module and/or a natural language understanding module. After the user terminal **201** recognizes the voice command of the user, text information corresponding to the recognized voice command may be transmitted to the intelligent server **300**. For example, the user terminal **201** may include a text-to-speech module. The user terminal **201** may receive text information from the intelligent server **300** and output the received text information as voice.

(71) The capsule database **330** may store information on relationships between a plurality of concepts and actions corresponding to a plurality of domains. A capsule according to an embodiment may include a plurality of action objects (or action information) and/or concept objects (or concept information) included in the plan. According to an embodiment, the capsule database **330** may store a plurality of capsules in the form of a concept action network (CAN). According to an embodiment, the plurality of capsules may be stored in a function registry included in the capsule database **330**.

(72) The capsule database **330** may include a strategy registry in which strategy information necessary for determining a plan corresponding to a voice input is stored. The strategy information may include reference information for determining one plan when there are a plurality of plans corresponding to the voice input. According to an embodiment, the capsule database **330** may include a follow up registry in which information on a subsequent action in a specified situation is stored. This is for the purpose of suggesting the subsequent action to the user. The subsequent action may include, for example, a subsequent utterance. According to an embodiment, the capsule database **330** may include a layout registry that stores layout information regarding information output through the user terminal **201**. According to an embodiment, the capsule database **330** may include a vocabulary registry in which vocabulary information included in the capsule information is stored. According to an embodiment, the capsule database **330** may include a dialog registry in which information regarding dialog (or interaction) with the user is stored. The capsule database **330** may update a stored object through a developer tool. The developer tool may include, for example, a function editor for updating an action object or a concept object. The developer tool may include a vocabulary editor for updating the vocabulary. The developer tool may include a strategy editor for generating and registering strategies for determining plans. The developer tool may include a dialog editor for generating dialog with the user. The developer tool may include a follow up editor that may edit follow-up utterances that activate subsequent goals and provide hints. The subsequent goal may be determined based on various parameters, such as a currently set goal, a user's preference, or an environmental condition. In an embodiment, the capsule database **330** may be implemented in the user terminal **201** as well.

(73) The execution engine **340** according to an embodiment may calculate a result by using the generated plan. The end user interface **350** may transmit the calculated result to the user terminal **201**. Accordingly, the user terminal **201** may receive the result and provide the received result to the user. The management platform **360** according to an embodiment may manage information

used in the intelligent server **300**. The big data platform **370** according to an embodiment may collect user data. The analytic platform **380** according to an embodiment may manage the quality of service (QoS) of the intelligent server **300**. For example, the analytic platform **380** may manage the components and processing speed (or efficiency) of the intelligent server **300**.

(74) The service server **400** according to an embodiment may provide a specified service (e.g., food order or hotel reservation) to the user terminal **201**. According to an embodiment, the service server **400** may be a server operated by a third party. The service server **400** according to an embodiment may provide, to the intelligent server **300**, information for generating a plan corresponding to the received voice input. The provided information may be stored in the capsule database **330**. In addition, the service server **400** may provide result information according to the plan to the intelligent server **300**. The service server **400** may communicate with the intelligent server **300** and/or the user terminal **201** through the network **299**. The service server **400** may communicate with the intelligent server **300** through a separate connection. Although the service server **400** is illustrated as one server in FIG. 2, embodiments of the present disclosure are not limited thereto. At least one of the respective services **401**, **402**, and **403** of the service server **400** may be implemented as a separate server.

(75) In the integrated intelligent system described above, the user terminal **201** may provide various intelligent services to the user in response to a user input. The user input may include, for example, an input via a physical button, a touch input, or a voice input.

(76) In an embodiment, the user terminal **201** may provide a speech recognition service through an intelligent app (or a speech recognition app) stored therein. In this case, for example, the user terminal **201** may recognize user utterance or voice input received through the microphone **270**, and provide a service corresponding to the recognized voice input to the user.

(77) In an embodiment, the user terminal **201** may perform a specified operation alone or together with the intelligent server **300** and/or the service server **400**, based on the received voice input. For example, the user terminal **201** may execute an app corresponding to the received voice input and perform a specified operation through the executed app.

(78) In an embodiment, when the user terminal **201** provides a service together with the intelligent server **300** and/or the service server **400**, the user terminal **201** may detect the user utterance by using the microphone **270** and generate a signal (or voice data) corresponding to the detected user utterance. The user terminal **201** may transmit the voice data to the intelligent server **300** by using the communication interface **290**.

(79) In response to the voice input received from the user terminal **201**, the intelligent server **300** according to an embodiment may generate a plan for performing a task corresponding to the voice input, or the result of performing one or more actions according to the plan. The plan may include, for example, a plurality of actions for performing the task corresponding to the voice input of the user and/or a plurality of concepts related to the plurality of actions. The concepts may define parameters input to the execution of the plurality of actions or result values output by the execution of the plurality of actions. The plan may include relation information between a plurality of actions and/or a plurality of concepts.

(80) The user terminal **201** according to an embodiment may receive the response by using the communication interface **290**. The user terminal **201** may output a voice signal generated in the user terminal **201** by using the speaker **255** so that the result is outputted as audio, or output an image generated in the user terminal **201** by using the display **260**.

(81) FIG. 3 is a diagram illustrating concepts and actions stored in a database, according to an embodiment.

(82) A capsule database (e.g., the capsule database **330**) of the intelligent server **300** may store capsules in the form of a concept action network (CAN). The capsule database may store an action for processing a task corresponding to a voice input of the user and a parameter necessary for the action in the form of the concept action network (CAN).

(83) The capsule database may store a plurality of capsules (capsule A **331** and capsule B **334**) corresponding to a plurality of domains (e.g., applications), respectively. According to an embodiment, one capsule (e.g., the capsule A **331**) may correspond to one domain (e.g., location, application). In addition, one capsule may correspond to a capsule of at least one service provider for performing a function for a domain related to the capsule (e.g., CP 1 **332**, CP 2 **333**, CP3 **335**, and/or CP4 **336**). According to an embodiment, one capsule may include at least one action **330a** and at least one concept **330b** for performing a specified function.

(84) The natural language platform **320** may generate a plan for performing a task corresponding to the voice input received by using the capsules stored in the capsule database **330**. For example, the planner module **325** of the natural language platform may generate a plan by using capsules stored in the capsule database. For example, a plan **337** may be generated by using actions **331a** and **332a** and concepts **331b** and **332b** of the capsule A **331** and an action **334a** and a concept **334b** of the capsule B **334**.

(85) FIG. 4 is a diagram illustrating a screen in which the user terminal processes a voice input received through the intelligent app, according to an embodiment.

(86) The user terminal **201** may execute an intelligent app to process the user input through the intelligent server **300**.

(87) According to an embodiment, if a specified voice input (e.g., wake up!) is recognized or an input is received via a hardware key (e.g., dedicated hardware key) or a software key, the user terminal **201** may execute the intelligent app to process the voice input on a first screen **210**. The user terminal **201** may, for example, execute the intelligent app when the schedule app is being executed. According to an embodiment, the user terminal **201** may display an object (e.g., icon) **211** corresponding to the intelligent app on the display **260**. According to an embodiment, the user terminal **201** may receive a voice input of a user utterance. For example, the user terminal **201** may receive a voice input saying "Tell me the schedule of the week!" According to an embodiment, the user terminal **201** may display a user interface (UI) **213** (e.g., input window) of the intelligent app in which text data of the received voice input is displayed on the display.

(88) According to an embodiment, on the second screen **215**, the user terminal **201** may display the result corresponding to the received voice input on the display. For example, the user terminal **201** may receive a plan corresponding to the received user input, and display "schedule of this week" on the display according to the plan.

(89) FIG. 5 illustrates a system for controlling a target device based on an utterance, according to an example.

(90) Referring to FIG. 5, a system **500** may include a first user device **501**, a second user device **502**, and a server device **511**. Each of the first user device **501** and the second user device **502** may be referred to as a listener device that receives utterance **590** of a user **599**, and may include components similar to those of the user terminal **201** of FIG. 2 or the electronic device **101** of FIG. 1. Each of the first user device **501** and the second user device **502** may include a voice assistant application (e.g., the client module **231** of FIG. 2). Each of the first user device **501** and the second user device **502** may be configured to receive the utterance **590** of the user **599** using voice receiving circuitry (e.g., the audio module **170** of FIG. 1), and transmit utterance data corresponding to the utterance **590** to the server device **511**. For example, each of the first user device **501** and the second user device **502** may be configured to transmit utterance data to the server device **511** through a network such as the Internet.

(91) The target device may be referred to as a device to be controlled by the utterance **590**. According to an embodiment, the target device may be one of the first user device **501** and the second user device **502** that have received the utterance **590**. In examples of the present disclosure, by determining one of the devices that have received (e.g., successfully received) the utterance **590** as the target device, the target device may be determined to be matched with the intent of the user **599**. In an example, the target device may be configured to receive control data from the server

device **511** through a network such as the Internet and perform one or more operations (e.g., sequential operations) according to the control data.

(92) The server device **511** may include at least one server device. For example, the server device **511** may include a first server **512** and a second server **513**. The server device **511** may be configured to receive utterance data from each of the first user device **501** and the second user device **502**, and process the utterance data. For example, the first server **512** may correspond to the intelligent server **300** of FIG. 2. The second server **513** may include a database of an external electronic device. The “external electronic device” may refer to an electronic device located outside the server device **511**. For example, information on the first user device **501** and/or the second user device **502** may be stored in the database of the second server **513**. The second server **513** may be referred to as an Internet-of-things (IoT) server. For example, the second server **513** may store information about the external electronic device (e.g., an identifier of the external electronic device, group information, or the like), and may include components for controlling the external electronic device. The first server **512** may determine the intent of the user **599** included in the received utterance data by processing the received utterance data. When the intent of the user **599** is to control an external device (e.g., the target device), the first server **512** may use data of the second server **513** to identify the target device to be controlled, and may control the target device so that the identified target device performs an operation according to the intent. Although the first server **512** and the second server **513** are illustrated as separate components in FIG. 5, the first server **512** and the second server **513** may be implemented as one server.

(93) The configuration of the system **500** illustrated in FIG. 5 is only an example, and embodiments of the present disclosure are not limited thereto. Various methods for controlling the target device may be included in the embodiments of the present disclosure.

(94) In an example, the utterance data transmitted by each of the first user device **501** and the second user device **502** to the server device **511** may be in any type of file format suitable for voice recording. In this case, the server device **511** may determine the intent of the user **599** for the utterance data by deploying speech recognition and natural language analysis of the utterance data. In an example, the utterance data transmitted by each of the first user device **501** and the second user device **502** to the server device **511** may include the recognition result of speech corresponding to the utterance **590**. In this case, each of the first user device **501** and the second user device **502** may perform automatic speech recognition on the utterance **590** and transmit the result of the automatic speech recognition to the server device **511** as the utterance data. In this case, the server device **511** may determine the intent of the user **599** for the utterance data by deploying just the natural language analysis of the utterance data.

(95) In an example, each of the first user device **501** and the second user device **502** may be configured to perform automatic speech recognition and natural language understanding. Each of the first user device **501** and the second user device **502** may be configured to directly identify the intent of the user **599** from the utterance **590**. In this case, each of the first user device **501** and the second user device **502** may use information stored in the second server **513** or information shared between the first user device **501** and the second user device **502** to identify the target device, and may control the target device according to the intent of the user **599**.

(96) In an example, the system **500** may not include the server device **511**. For example, each of the first user device **501** and the second user device **502** may be configured to perform all of the operations of the server device **511** described above.

(97) The examples described above with reference to FIG. 5 are various examples of the system capable of controlling the target device based on the utterance, and embodiments of the present disclosure are not limited thereto. It should be understood to those skilled in the art that the control methods of the present disclosure described below may be carried out using the examples described above with reference to FIG. 5, as well as other embodiments.

(98) FIG. 6 illustrates a multi-device environment according to an example.

(99) Referring to FIG. 6, a multi-device environment **600** according to an embodiment may include at least one listener device and at least one target device (e.g., the device to be controlled).

(100) For example, a smart watch **601**, a mobile phone **602**, and an artificial intelligence (AI) speaker **603** may correspond to the user devices **501** and **502** of FIG. 5. The user **699** may control an electronic device using a voice assistant application provided in the smart watch **601**, the mobile phone **602**, or the AI speaker **603**. For example, the user **699** may start the execution of the voice assistant by uttering a wake-up utterance or by generating a user input to a particular device (e.g., button input or touch input). The user can then control the target device by performing a control utterance.

(101) In the example of FIG. 6, a first light **621**, a standing lamp **622**, and a TV **623** are assumed to be located in the living room **681**, and a second light **624**, a third light **625**, and a refrigerator **626** may be assumed to be located in the kitchen **682**. In an example, the user **699** may use the voice assistant of the mobile phone **602** to execute a voice command. For example, the utterance of the user **699** may be as follows: "Turn on the living room light." Since the first light **621** and the standing lamp **622** are located in the living room **681**, the voice assistant of the mobile phone **602** may turn on the first light **621** and the standing lamp **622**. In this case, since the type (e.g., light) and location (e.g., living room) of the target device are specified by the utterance of the user **699**, the voice assistant may identify the target device from the utterance.

(102) In an example, the user **699** may want to control the device that is receiving the utterance via voice control. In this case, the utterance of the user **699** may not include information on the name or location of the device to be controlled. The user **699** directly controls the device that receives the utterance, and the user **699** may determine that separate information for identifying a target device is unnecessary. In this case, it is necessary to identify which device of the smart watch **601**, the mobile phone **602**, and the AI speaker **603** capable of receiving the utterance of the user **699** is the target of the utterance of the user **699**.

(103) According to embodiments of the present disclosure, a device capable of performing a function corresponding to the control utterance may be determined as the target device, among the devices that have received the utterance. In this case, even if the user **699** does not mention the name of the target device to be controlled, the target device matching the intent of the user **699** may be controlled based on the utterance. For example, context information about devices that have received the wake-up utterance may be collected, and the target device capable of performing the function corresponding to the control utterance may be determined based on the context information.

(104) Hereinafter, methods for identifying the target device may be described in detail with reference to FIGS. 7 to 12.

(105) FIG. 7 illustrates a block diagram of an electronic device according to an example.

(106) Referring to FIG. 7, according to an embodiment, an electronic device **701** may include a processor **720** (e.g., the processor **120** of FIG. 1), a memory **730** (e.g., the memory **130** of FIG. 1), and/or a communication circuitry **790** (e.g., the communication module **190** of FIG. 1). For example, the electronic device **701** may further include audio circuitry **750** (e.g., the audio module **170** of FIG. 1), and may further include other components not shown in FIG. 7. For example, the electronic device **701** may further include at least some components of the electronic device **101** of FIG. 1.

(107) In examples of the present disclosure, the electronic device **701** may be referred to as a device for identifying and/or determining the target device. For example, if identification and/or determination of the target device is performed in a server device (e.g., the server device **511** of FIG. 5), the electronic device **701** may be referred to as the server device. For example, if identification and/or determination of the target device is performed in a user device (e.g., the first user device **501** and/or the second user device **502** of FIG. 5), the electronic device **701** may be referred to as the user device. The fact that, after the target device is identified, control of the target

device may be performed using another device is as described above. Accordingly, the electronic device **701** may directly control the target device or may indirectly control the target device through another device.

(108) The processor **720** may be electrically, operatively, or functionally connected to the memory **730**, the communication circuitry **790**, and/or the audio circuitry **750**. The memory **730** may store instructions. When the instructions are executed by the processor **720**, the instructions may cause the electronic device **701** to perform various operations.

(109) According to an embodiment, the electronic device **701** may be configured to acquire state information indicating states of a plurality of external electronic devices that have received a wake-up utterance from the plurality of external electronic devices. For example, each of the plurality of external electronic devices may be configured to receive the wake-up utterance using their audio receiving circuitry. For example, the state information may include active service information about each of the plurality of external electronic devices. For example, the active service may be a service currently running in each of the plurality of external electronic devices.

(110) According to an embodiment, the electronic device **701** may be configured to acquire control data corresponding to a control utterance subsequent to the wake-up utterance. For example, the control utterance may indicate control of a first function. For example, the control data may not include identification information (e.g., name) about the target device. The electronic device **701** may be configured to identify an intent corresponding to the control data based on speech recognition of the control data and to identify the first function corresponding to the intent.

(111) According to an embodiment, the electronic device **701** may be configured to determine availability of the first function for at least some of the plurality of external electronic devices by using the state information. For example, the electronic device **701** may be configured to identify functions corresponding to the active service indicated by the active service information, and to determine the availability by comparing the identified functions with the first function.

(112) According to an embodiment, the electronic device **701** may be configured to determine the target device to perform the first function from the plurality of external electronic devices, based on the availability. For example, the electronic device **701** may be configured to identify a first device group supporting the first function among a plurality of external electronic devices. The electronic device **701** may be configured to determine the availability of the first function in the first device group by using the state information, and to determine the target device based on the availability. For example, if it is determined that the first function is available in a first external electronic device and a second external electronic device, the electronic device **701** may be configured to determine, from the first external electronic device and the second external electronic device, the device with high reception volume or reception sensitivity of the wake-up utterance as the target device. For an example, the electronic device **701** may be configured to determine the device with higher frequency of use as the target device from the first external electronic device and the second external electronic device.

(113) According to an embodiment, the electronic device **701** may be configured to control the target device so as to perform the first function. For example, the electronic device **701** may control the target device by directly transmitting information indicating performance of the first function to the target device or indirectly transmitting through another device.

(114) FIG. **8** illustrates a system for controlling a listener device according to an example.

(115) Referring to FIG. **8**, the system **800** may include various modules for controlling at least one of a plurality of listener devices **805** based on the utterance of the user **899**. The term “module” in FIG. **8** may refer to a software module or a hardware module. If the module is a software module, the module may be implemented by executing instructions by a processor.

(116) In an example, a natural language processing module **810** may be included in the first server **512** of FIG. **5**, and a dispatcher module **820** and a device control module **830** may be included in the second server **513** of FIG. **5**. In an example, each of the plurality of listener devices **805** may

include the natural language processing module **810**, and the server device **511** of FIG. 5 may include the dispatcher module **820** and the device control module **830**. In an example, each of the plurality of listener devices **805** may include the natural language processing module **810**, the dispatcher module **820**, and the device control module **830**. As described above with reference to FIG. 7, the electronic device **701** of FIG. 7 may be referred to as an electronic device including at least a device determination module **822**.

(117) Each of the plurality of listener devices **805** is a device in which a voice assistant application is installed, and may receive the utterance of the user **899** and transmit utterance data corresponding to the utterance to a server device (e.g., the first server **512** in FIG. 5). Hereinafter, for convenience of description, operations of the first listener device **801** are described, but the second listener device **802** and the third listener device **803** may also be configured to perform the same or similar operations.

(118) For example, the first listener device **801** may receive a wake-up utterance **890** from the user **899**. The first listener device **801** may activate the voice assistant application and activate a microphone (e.g., the audio circuitry **750** of FIG. 7), in response to receiving the wake-up utterance **890**. The first listener device **801** may transmit state information about the first listener device **801** to the dispatcher module **820** in response to receiving the wake-up utterance **890**. The state information about the first listener device **801** may include information on the state of at least one function currently being executed in the first listener device **801**. For example, the state information may include information on a service currently activated in the first listener device **801**. If music is being played in the first listener device **801**, the state information about the first listener device **801** may indicate that the music playback service is activated. If there is no service currently activated in the first listener device **801**, the state information about the first listener device **801** may indicate that there is no service currently activated. In an example, the state information may further include device type information (e.g., mobile, speaker, TV, or the like) and/or a winner device indicator (described below).

(119) In the example of FIG. 8, the plurality of listener devices **805** may be assumed to have received the wake-up utterance **890**. In examples of the present disclosure, “reception of an utterance” may refer to successful reception of the utterance. “Successful reception” may mean, for example, when the speech of the utterance is recognizable, when the signal-to-noise ratio (SNR) of the utterance is greater than or equal to a specified value, or when the volume of the utterance is greater than or equal to a specified value.

(120) As described above, each of the plurality of listener devices **805** may transmit state information to the dispatcher module **820** in response to the wake-up utterance **890**. For example, if the dispatcher module **820** is included in the server, each of the plurality of listener devices **805** may transmit state information to the dispatcher module **820** through a network such as the Internet. For another example, if the dispatcher module **820** is included in each of the listener devices, each of the plurality of listener devices **805** may share state information with each other by transmitting (e.g., broadcasting) its own state information through a local area network (e.g., network(s) based on WiFi, Bluetooth, NAAN, and/or UWB).

(121) The winner device may be determined from the plurality of listener devices **805**. For example, each of the plurality of listener devices **805** may transmit reception information (e.g., reception volume and/or SNR) for the wake-up utterance **890** to the dispatcher module **820**. The dispatcher module **820** may determine the winner device from the plurality of listener devices **805** by using the reception information for the wake-up utterance **890**. The dispatcher module **820** may determine the device with the highest reception volume or the device with the highest SNR as the winner device by using the reception information. For example, each of the plurality of listener devices **805** may share reception information for the wake-up utterance **890** with each other. Each of the plurality of listener devices **805** may determine the winner device by using the reception information. For example, each of the plurality of listener devices **805** may determine the device

with the highest reception volume or the device with the highest SNR as the winner device. Each of the plurality of listener devices **805** may be configured to transmit state information including information about the winner device (e.g., winner device indicator), if the winner device is determined by the plurality of listener devices **805**.

(122) In the example of FIG. **8**, the plurality of listener devices **805** may receive a control utterance **891** subsequent to the wake-up utterance **890**. The control utterance **891** is an utterance instructing control of a specific function, and in the example of FIG. **8**, the control utterance **891** may not include information on the name of the target device to be controlled. The control utterance **981** may be transmitted from the plurality of listener devices **805** to the natural language processing module **810**.

(123) The natural language processing module **810** may identify an intent of the user based on utterance data (e.g., data corresponding to the control utterance **981**) received from the plurality of listener devices **805**. For example, the natural language processing module **810** may correspond to the intelligent server **300** of FIG. **2** (e.g., the first server **512** of FIG. **5**). Although not shown in FIG. **8**, the natural language processing module **810** may further include a front end (e.g., the front end **310** of FIG. **2**).

(124) The natural language processing module **810** may generate text data from utterance data by using automatic speech recognition module **811** (e.g., the automatic speech recognition module **321** of FIG. **2**) to perform speech recognition on the utterance data. The natural language processing module **810** may use a natural language understanding module **812** (e.g., the natural language understanding module **323** of FIG. **2**) to identify the intent from the text data. For example, the natural language processing module **810** may identify the intent corresponding to the utterance (e.g., the control utterance **891**) by comparing a plurality of predefined intents with the text data. Further, the natural language processing module **810** may extract additional information from the utterance data. For example, the natural language processing module **810** may perform slot tagging or slot filling by extracting words (e.g., entities) included in the utterance data.

(125) The natural language processing module **810** may use a natural language generator module **813** (e.g., the natural language generator module **327** of FIG. **2**) to provide feedback (e.g., response) on the utterance (e.g., the control utterance **891**) to the user. For example, the natural language generator module **813** may generate the feedback to be provided to the user as a natural language expression. A text-to-speech module **814** (e.g., the text-to-speech module **329** of FIG. **2**) may convert the natural language expression generated by the natural language generator module **813** from text data to voice data. For example, the natural language processing module **810** may provide a response to the utterance to the user **899** through the first listener device **801** by transmitting the generated voice data to the first listener device **801**.

(126) The natural language processing module **810** may transmit the intent identified from the utterance of the user **899** (e.g., the control utterance **891**) to the dispatcher module **820**. The natural language processing module **810** may transmit the identified intent and the extracted additional information (e.g., entity) from the control utterance **891** to the dispatcher module **820**.

(127) According to an embodiment, the dispatcher module **820** may include a detector **821**, the device determination module **822**, and a context classifier **823**. The detector **821** may transmit information on a candidate device capable of performing the function corresponding to the control utterance **891** (e.g., the function corresponding to the intent of the control utterance **891**) to the device determination module **822**. For example, the detector **821** may transmit, to the device determination module **822**, a list of candidate devices capable of executing the function corresponding to the control utterance **891**, among devices associated with the user **899** in the plurality of listener devices **805** (e.g., electronic devices registered to the account of the user **899**). Table 1 shows an example of candidate device information if the control utterance **891** is “volume up”.

(128) TABLE-US-00001 TABLE 1 Utterance Intent Candidate Device Information Volume Up

Media-Volume-Up TV in Bedroom TV in Living Room Speaker 1 in Living Room Speaker 2 in Living Room Mobile Phone

(129) The context classifier **823** may use the state information about the plurality of listener devices **805** that have received the wake-up utterance **890** to determine availability of functions of each of the plurality of listener devices **805**. For example, if the state information about the third listener device **803** indicates media is being played, the context classifier **823** may identify functions that may be executed during media playback (e.g., adjust volume, stop playback, and so on) as available functions. The context classifier **823** may include a database of functions mapped to each service, and may determine the availability of functions mapped to the service based on the state of the service received from listener devices. The context classifier **823** may transmit the availability of each function identified based on the state information to the device determination module **822**.

(130) Table 2 below shows state information according to an example.

(131) TABLE-US-00002

TABLE 2	Device	Device Type	Winner	Device Active	Service	First
Mobile Phone	TRUE	NONE	Listener Device	Second	TV in Living	FALSE
NONE	Listener	Room	Device	Third	Speaker in	FALSE
MUSIC	Listener	Living Room	PLAYBACK	Device		

(132) The context classifier **823** may identify available functions mapped to active services of the third listener device **803**. Table 3 below shows available functions according to an example.

(133) TABLE-US-00003

TABLE 3	Device	Type	Active Service	Available Functions
Speaker in Living	MUSIC	PLAY	Media-Volume-Up	Room Media-Volume-Down Media-Stop Etc.

(134) According to an embodiment, the device determination module **822** may determine the target device from the plurality of listener devices **805** based on availability. The device determination module **822** may determine a device in which the function corresponding to the intent is available, as the target device from the candidate devices indicated by the candidate device information. In the examples of Tables 1 to 3, the device determination module **822** may determine the third listener device **803** as the target device, because the “Media-Volume-Up” function is available in the third listener device **803**, from the candidate devices of Table 1.

(135) If the target device is determined, the dispatcher module **820** may transmit, to the device control module **830**, identification information about the determined target device and the intent identified from the control utterance **891**. The device control module **830** may transmit, to the target device, information for performing an operation corresponding to the intent. In the example of FIG. 8, the device control module **830** may increase the playback volume of the third listener device **803** by transmitting, to the third listener device **803**, information (e.g., link information specified for the corresponding function) for performing the operation corresponding to “Media-Volume-Up”.

(136) In an example, a plurality of electronic devices in the listener devices may be in a state in which the function corresponding to the intent of the control utterance **891** is available. For example, the second listener device **802** and the third listener device **803** may be in states in which “Media-Volume-Up” is available. For example, the device determination module **822** may determine the winner device indicated by the state information as the target device. For example, the device determination module **822** may determine, from the second listener device **802** and the third listener device **803**, the device with higher reception volume or the device with higher SNR of the wake-up utterance **890** as the target device. For an example, the device determination module **822** may determine both the second listener device **802** and the third listener device **803** as target devices to cause the second listener device **802** and the third listener device **803** to perform the function corresponding to the control utterance **891**. For example, the device determination module **822** may ask the user **899** which one of the second listener device **802** and the third listener device **803** wants to be controlled. In this case, the device determination module **822** may ask the user **899** using the winner device of the plurality of listener devices **805**. The device determination module **822** may determine the target device based on the response of the user **899**.

(137) In an example, there may be none of the listener devices in which the function corresponding to the intent of the control utterance **891** is available. In this case, the device determination module **822** may determine the target device based on history information. The device determination module **822** may identify a device included in the candidate devices in the plurality of listener devices **805**, and determine a listener device with the highest voice-based control frequency as the target device from the identified listener devices. In another example, the device determination module **822** may identify a device included in the candidate devices, and may inquire of the user **899** which one of the identified devices he or she wants to be controlled. The device determination module **822** may determine the target device based on the response of the user **899** to the inquiry.

(138) FIG. **9** is a signal flow diagram illustrating a method for controlling a target device according to an example.

(139) Referring to FIGS. **7**, **8**, and **9**, the electronic device **701** may be a device separate from the listener device (e.g., the server device **511** of FIG. **5**). For example, the detector **821**, the device determination module **822**, the context classifier **823**, and the device control module **830** may be referred to as components of the electronic device **701**. According to an embodiment, the electronic device **701** may determine a target device from the listener devices.

(140) In operation **901** of a signal flow diagram **900**, the first listener device **801** and the second listener device **802** may receive a wake-up utterance. In operation **903**, the first listener device **801** may transmit state information about the first listener device **801** to the context classifier **823** in response to receiving the wake-up utterance. The state information may include information regarding an active service of the first listener device **801**, type information, and/or a winner device indicator, as described above. In operation **905**, the second listener device **802** may transmit state information about the second listener device **802** to the context classifier **823** in response to receiving the wake-up utterance. The state information may include information regarding an active service of the second listener device **802**, type information, and/or a winner device indicator.

(141) In operation **907**, the first listener device **801** and the second listener device **802** may receive a control utterance. For example, the control utterance may indicate control of the first function and may not include the name of the target device. The first listener device **801** and the second listener device **802** may transmit data corresponding to the control utterance to a speech recognition module (e.g., the natural language processing module **810** of FIG. **8**) in response to receiving the control utterance.

(142) In operation **909**, the detector **821** may identify one or more candidate devices. For example, the detector **821** may receive an intent (e.g., control target function) corresponding to the control utterance from a speech recognition module (not shown), and identify candidate devices by using the intent. The detector **821** may identify, as the candidate devices, devices associated with the first listener device **801** and the second listener device **802** (e.g., registered with a user account of the first listener device **801** and the second listener device **802**) and supporting the function corresponding to the intent of the control utterance.

(143) In operation **911**, the detector **821** may transmit candidate device information to the device determination module **822**. In operation **913**, the context classifier **823** may transmit state information corresponding function information to the device determination module **822**. The context classifier **823** may transmit, to the device determination module **822**, the state information corresponding function information, which includes information on available functions corresponding to the state information (e.g., active service information) about the first listener device **801**. The context classifier **823** may transmit, to the device determination module **822**, the state information corresponding function information, which includes information on available functions corresponding to the state information (e.g., active service information) about the second listener device **802**. In the example of FIG. **9**, operation **913** has been illustrated as being performed after operation **911**. However, examples of the present disclosure are not limited thereto. Operation **913** may be performed at any time after the reception of the control utterance (e.g., operation **907**).

(144) In operation **915**, the device determination module **822** may identify the target device. For example, the device determination module **822** may identify the listener device in which state information corresponding function information indicates that the function corresponding to the control utterance is available, as the target device selected from the candidate devices indicated by the candidate device information. For example, the device determination module **822** may identify the device in which the function corresponding to the control utterance is available, as the target device selected from devices listed in the candidate device information and also are listener devices. For example, in the example of FIG. **9**, the device determination module **822** may identify the first listener device **801** as the target device.

(145) In operation **917**, the device determination module **822** may transmit target device information to the device control module **830**. For example, the device determination module **822** may transmit identification information about the first listener device **801** to the device control module **830**.

(146) In operation **919**, the device control module **830** may transmit a control utterance corresponding control command (a command corresponding to the intent of the control utterance) to the first listener device **801**. The device control module **830** may receive the target device information from the device determination module **822**, and may receive the control command from the speech recognition module (not shown). In another example, the device control module **830** may receive the target device information and the control command from the device determination module **822**.

(147) FIG. **10** is a signal flow diagram illustrating a method for controlling a target device according to an example.

(148) Referring to FIGS. **7**, **8**, and **10**, the electronic device **701** may be a listener device. For example, the detector **821**, the device determination module **822**, the context classifier **823**, and the device control module **830** may be referred to as components of the electronic device **701**. According to an embodiment, the electronic device **701** may determine a target device from the listener devices.

(149) In operation **1001** of a signal flow diagram **1000**, the first listener device **801**, the second listener device **802**, and the third listener device **803** may receive a wake-up utterance. In operation **1003**, the first listener device **801**, the second listener device **802**, and the third listener device **803** may share state information in response to receiving the wake-up utterance. For example, each of the first listener device **801**, the second listener device **802**, and the third listener device **803** may share state information by transmitting (e.g., broadcasting) its own state information through the local area network. The state information may include information regarding active service, type information, and/or wake-up utterance reception information (e.g., reception strength and/or SNR) for each listener device.

(150) In operation **1005**, the first listener device **801**, the second listener device **802**, and the third listener device **803** may receive a control utterance. For example, the control utterance may indicate control of the first function and may not include the name of the target device.

(151) In operation **1007**, each of the first listener device **801**, the second listener device **802**, and the third listener device **803** may identify the target device. For example, each listener device may determine based on its current active service whether it can perform the function corresponding to the control utterance. Each listener device may share its current availability of the function corresponding to the control utterance among the listener devices by transmitting (e.g., broadcasting) information indicating the availability. Each listener device may identify the target device based on the availability. If a single listener device is identified as the only target device (e.g., if information received from another listener device indicates that the function corresponding to the control utterance is not available), the listener device may execute the control function as the target device. For example, if the second listener device **802** is identified as the target device, in operation **1009**, the second listener device **802** may execute the control function.

(152) FIG. 11 illustrates a control environment of a target device according to an example.

(153) Referring to FIG. 11, in a control environment 1100, a user 1199 may want to control another device via voice command. The user 1199 may call the voice assistant through a wake-up utterance (e.g., utterance of a wake-up word). In the example of FIG. 11, it may be assumed that, among a first electronic device 1111, a second electronic device 1112, and a third electronic device 1113, the second electronic device 1112 and the third electronic device 1113 have received the utterance. In the examples of the present disclosure, the first electronic device 1111 that has failed to receive the wake-up utterance may be excluded as a candidate for being a target device of a subsequent control utterance. By determining only the devices that have directly received the wake-up utterance as potential target devices, the electronic devices may better execute the intent of the user.

(154) The user 1199 may speak the control utterance following the wake-up utterance. For example, the user 1199 may say "Volume up". In the example of FIG. 11, the second electronic device 1112 may be playing music, and the third electronic device 1113 may be on standby. In this case, the second electronic device 1112 may be in the state in which media volume control is available, and the third electronic device 1113 may be in the state in which media volume control is not available. According to embodiments of the present disclosure, from devices that have received the wake-up utterance, the device in which the function corresponding to the control utterance is available may be determined as the target device. For example, the second electronic device 1112 may be determined as the target device. In this case, the second electronic device 1112 may perform the operation corresponding to the control utterance of the user. The second electronic device 1112 may provide feedback to the user 1199 after performing the operation corresponding to the control utterance.

(155) FIG. 12 is a flowchart illustrating a method for controlling a target device according to an example.

(156) Referring to FIGS. 7 and 12, according to an embodiment, the electronic device 701 may determine a target device based on availability of a function corresponding to a control utterance.

(157) In operation 1205, the electronic device 701 may acquire state information from a plurality of external electronic devices. Each of the plurality of external electronic devices may be configured to transmit the state information in response to reception of a wake-up utterance. For example, each of the plurality of external electronic devices may be configured to receive the wake-up utterance using audio receiving circuitry. The state information may include active service information, device type information, and/or winner device information for each of the plurality of external electronic devices. It may be assumed that each of the plurality of external electronic devices has successfully received the wake-up utterance.

(158) In operation 1210, the electronic device 701 may acquire control data corresponding to the control utterance. For example, the electronic device 701 may acquire control data corresponding to the control utterance by receiving the control data from a plurality of external electronic devices. The control data may be data acquired through voice recognition for the control utterance. The control data may not include identification information (e.g., name) about the target device. As an example, the electronic device 701 may be configured to identify an intent corresponding to the control data based on speech recognition of the control data and to identify a first function corresponding to the intent.

(159) In operation 1215, the electronic device 701 may determine availability of the function (hereinafter, referred to as the control function) corresponding to the control data by using the state information. The electronic device 701 may identify state information about each of the plurality of external electronic devices and a function mapped to a service indicated by the state information. The electronic device 701 may identify the function mapped to the service (e.g., active service) indicated by the state information as an available function. For example, if functions mapped to services activated in the first external electronic device include the control function, the electronic device 701 may determine that the control function is available in the first external electronic

device. For example, the active service may correspond to a service currently running in each of the plurality of external electronic devices.

(160) In operation **1220**, the electronic device **701** may determine the target device from a plurality of external electronic devices based on the availability. For example, the electronic device **701** may determine an external electronic device capable of performing the control function as the target device. In another example, if the control function is available in a plurality of external electronic devices, the electronic device **701** may determine the target device according to one of the methods described above with reference to FIG. 8.

(161) For example, the determining of the target device (operation **1220**) may include identifying a first device group supporting the first function among external electronic devices associated with accounts of the plurality of external electronic devices, identifying a second device group consisting of devices belonging to the first device group among the plurality of external electronic devices, determining availability of the first function of external electronic devices in the second device group by using the state information, and determining the target device based on the availability.

(162) For example, the determining of the target device (operation **1220**) may include determining that the first function is available in a first external electronic device and a second external electronic device in the plurality of external electronic devices, based on the state information, and determining, in the first external electronic device and the second external electronic device, the device with a high reception volume or reception sensitivity of the wake-up utterance as the target device.

(163) For example, the determining of the target device (operation **1220**) may include determining that the first function is available in a first external electronic device and a second external electronic device in the plurality of external electronic devices, based on the state information, and determining, in the first external electronic device and the second external electronic device, the device with a high frequency of use as the target device.

(164) In operation **1225**, the electronic device **701** may control the target device so that the target device performs the function corresponding to the control utterance. For example, the electronic device **701** may control the determined target device by transmitting, to the target device, information instructing control of the function corresponding to the control data.

(165) Certain of the above-described embodiments of the present disclosure can be implemented in hardware, firmware or via the execution of software or computer code that can be stored in a recording medium such as a CD ROM, a Digital Versatile Disc (DVD), a magnetic tape, a RAM, a floppy disk, a hard disk, or a magneto-optical disk or computer code downloaded over a network originally stored on a remote recording medium or a non-transitory machine readable medium and to be stored on a local recording medium, so that the methods described herein can be rendered via such software that is stored on the recording medium using a general purpose computer, or a special processor or in programmable or dedicated hardware, such as an ASIC or FPGA. As would be understood in the art, the computer, the processor, microprocessor controller or the programmable hardware include memory components, e.g., RAM, ROM, Flash, etc. that may store or receive software or computer code that when accessed and executed by the computer, processor or hardware implement the processing methods described herein.

(166) While the present disclosure has been shown and described with reference to various embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the present disclosure as defined by the appended claims and their equivalents.

Claims

1. An electronic device comprising: a microphone; communication circuitry; a processor; and a memory that stores instructions, wherein the instructions, when executed by the processor, cause the electronic device to: obtain a wake-up utterance using the microphone; broadcast, based on obtaining the wake-up utterance, state information of the electronic device indicating at least one service being currently executed by the electronic device; obtain, from a plurality of external electronic devices that have received the wake-up utterance, state information indicating currently executed services of the plurality of external electronic devices; obtain control data corresponding to a control utterance subsequent to the wake-up utterance, the control utterance instructing control of a first function; identify a first group of devices, among the electronic device and the plurality of external electronic devices, registered to an account of the electronic device; determine availability of the first function for the first group of devices by identifying available functions supported by currently executed services, the currently executed services being indicated by the state information of the electronic device and the plurality of external electronic devices; determine a target device to perform the first function among the first group of devices, based on the availability; and control the target device to perform the first function.
2. The electronic device of claim 1, wherein the instructions, when executed by the processor, cause the electronic device to: determine the availability by comparing the available functions with the first function.
3. The electronic device of claim 1, wherein the instructions, when executed by the processor, cause the electronic device to: identify an intent corresponding to the control data based on speech recognition of the control data; and identify the first function as corresponding to the intent.
4. The electronic device of claim 1, wherein a name of the target device is absent in the control data.
5. The electronic device of claim 1, wherein the instructions, when executed by the processor, cause the electronic device to: determine that the first function is available in a first external electronic device and a second external electronic device among the first group of devices, based on the state information; and determine, from the first external electronic device and the second external electronic device, a device with a high reception volume or reception sensitivity of the wake-up utterance as the target device.
6. The electronic device of claim 1, wherein the instructions, when executed by the processor, cause the electronic device to: determine that the first function is available in a first external electronic device and a second external electronic device among the first group of devices, based on the state information; and determine, from the first external electronic device and the second external electronic device, a device with a high frequency of use as the target device.
7. The electronic device of claim 1, wherein each of the plurality of external electronic devices is configured to receive the wake-up utterance using audio receiving circuitry.
8. A method for controlling a target device of an electronic device, the method comprising: obtaining a wake-up: broadcasting, based on obtaining the wake-up utterance, state information of the electronic device indicating at least one service being currently executed by the electronic device; obtaining, from a plurality of external electronic devices that have received a wake-up utterance, state information indicating states of the plurality of external electronic devices; obtaining control data corresponding to a control utterance subsequent to the wake-up utterance, the control utterance instructing control of a first function; identifying a first group of devices, among the electronic device and the plurality of external electronic devices, registered to an account of the electronic device determining availability of the first function for the first group of devices by identifying available functions supported by currently executed services, the currently executed services being indicated by the state information of the electronic device and the plurality of external electronic devices; determining a target device to perform the first function among the first group of devices, based on the availability; and controlling the target device to perform the

first function.

9. The method of claim 8, wherein the determining availability includes: determining the availability by comparing the available functions with the first function.

10. The method of claim 8, further comprising: identifying an intent corresponding to the control data based on speech recognition of the control data; and identifying the first function as corresponding to the intent.

11. The method of claim 8, wherein a name of the target device is absent in the control data.

12. The method of claim 8, wherein the determining of the target device includes: determining that the first function is available in a first external electronic device and a second external electronic device among the first group of devices, based on the state information; and determining, from the first external electronic device and the second external electronic device, a device with a high reception volume or reception sensitivity of the wake-up utterance as the target device.

13. The method of claim 8, wherein the determining of the target device includes: determining that the first function is available in a first external electronic device and a second external electronic device among the first group of devices, based on the state information; and determining, from the first external electronic device and the second external electronic device, a device with a high frequency of use as the target device.

14. The method of claim 8, wherein each of the plurality of external electronic devices is configured to receive the wake-up utterance using audio receiving circuitry.
